Chapter 10 — Electrochemistry

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 232–235

Q.1 — Multiple Choice Questions (with correct option and reason)

I. The activity series of metals arranges metals in order of their:

(a) Atomic mass
(b) Density
(c) Ease of oxidation
(d) Ease of reduction

Answer: (c) Ease of oxidation — The activity series ranks metals by their tendency to lose electrons (be oxidized).

II. Which metal would most readily displace hydrogen gas from dilute acids?

(a) Copper (Cu)
(b) Silver (Ag)
(c) Magnesium (Mg)
(d) Platinum (Pt)

Answer: (c) Magnesium (Mg) — Mg is above hydrogen in the activity series and is readily oxidized, displacing H_2.

III. The salt bridge allows transfer of ---------- in a Zn-Cu voltaic cell.

(a) Zn^{2+} ions
(b) SO_4^{2-} ions
(c) Both
(d) None of these

Answer: (c) Both — The salt bridge permits migration of ions (anions toward the anode, cations toward the cathode) to maintain electrical neutrality.

The salt bridge actually carries its own inert ions (e.g. K^+/NO_3^-); the option "Both" is taken to mean ions of both signs.

IV. If a Zn-Cu galvanic cell works ideally after complete discharge, both compartments will have:

(a) $CuSO_4$ solution
(b) $ZnSO_4$ solution
(c) Cu^{2+} ions
(d) Zn solid

Answer: (b) $ZnSO_4$ solution — Cu^{2+} is fully reduced to Cu while Zn dissolves to Zn^{2+}, leaving $ZnSO_4$ solution in both compartments.

V. Which half-reaction represents a reduction process?

(a) $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$
(b) $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$
(c) $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$
(d) $2H^+(aq) + 2e^- \rightarrow H_2(g)$

Answer: (c) $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$ — Reduction is gain of electrons; Cl_2 gains electrons to form Cl^-.

Option (d), $2H^+ + 2e^- \rightarrow H_2$, is also a reduction; the question has two correct reduction half-reactions.

VI. Which half-reaction represents the oxidation process in the disproportionation of Cu^+?

(a) $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
(b) $Cu^+(aq) + e^- \rightarrow Cu(s)$
(c) $Cu^+(aq) \rightarrow Cu^{2+}(aq) + e^-$
(d) $2Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$

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Answer: (c) $Cu^{+}(aq) \rightarrow Cu^{2+}(aq) + e^{-}$ — Oxidation is loss of electrons: $Cu^{+} \rightarrow Cu^{2+} + e^{-}$.

VII. If a salt bridge is not used between two half-cells in a galvanic cell, the voltage:

(a) Decreases slowly
(b) Decreases rapidly
(c) Does not change
(d) Drops to zero

Answer: (d) Drops to zero — Without a salt bridge charge builds up and the circuit is incomplete, so the voltage drops to zero.

VIII. In electrolysis, a charge of 96,500 C corresponds to:

(a) Mass number of the ion
(b) One mole of electrons being transferred
(c) Avogadro's number of ions being discharged
(d) Standard electrode potential of the metal ion

Answer: (b) One mole of electrons being transferred — 96,500 C is one faraday, the charge of one mole of electrons.

IX. Experimental determination of Avogadro's number through electrolysis typically involves measuring:

(a) The current and voltage applied.
(b) The mass of substance deposited or liberated by a known charge.
(c) The conductivity of the electrolytic solution.
(d) The temperature changes during electrolysis.

Answer: (b) The mass of substance deposited or liberated by a known charge. — Mass deposited by a known charge gives the charge per mole of electrons (F), and $N_a = F/e$.

X. The principle of measuring DO by Winkler's method is based on:

(a) Iodimetry
(b) Iodometry
(c) Acid-base titration
(d) Complexometry

Answer: (b) Iodometry — The iodine liberated is titrated with thiosulfate — an iodometric method.

XI. A positive standard electrode potential (E°) of a metal ion/metal half-cell indicates that:

(a) The metal is a strong reducing agent.
(b) The metal ion is readily oxidized.
(c) The metal ion is readily reduced.
(d) The metal will readily displace hydrogen from dilute acids.

Answer: (c) The metal ion is readily reduced. — A more positive E° means a greater tendency for the ion to gain electrons (be reduced).

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XII. Which change would typically increase the rate of electrolysis?

(a) Decreasing the concentration of the electrolyte

(b) Increasing the distance between the electrodes

(c) Decreasing the surface area of the electrodes

(d) Increasing the current passed through the electrolytic cell

Answer: (d) Increasing the current passed through the electrolytic cell — A larger current passes more charge per second, increasing the rate of electrolysis.

Q.2 — Short Answer Questions

a. How and why is an electrical double layer formed?

When a metal is dipped in a solution of its ions, metal atoms may leave the surface as ions (leaving electrons on the metal) or ions may deposit on the metal. This charge separation attracts oppositely charged ions from solution, forming an electrical double layer at the metal–solution interface.

b. Why is the electrode potential of Cu called a reduction potential?

By convention, electrode potentials are written for the reduction half-reaction ($Cu^{2+} + 2e^- \rightarrow Cu$). The tendency of Cu^{2+} to gain electrons is measured relative to SHE, so it is called the reduction potential.

c. What are the advantages of a salt bridge in a galvanic cell?

It completes the electrical circuit, maintains electrical neutrality in the half-cells by ion migration, prevents mixing of the two solutions, and minimises the liquid-junction potential.

d. How can we predict the feasibility of a chemical reaction using cell voltage?

If $E°_{cell}$ is positive, $\Delta G° = -nFE°_{cell}$ is negative and the reaction is spontaneous (feasible) under standard conditions. A negative $E°_{cell}$ indicates a non-spontaneous reaction.

e. During electrolysis of aqueous NaCl, why is Na not liberated at the cathode?

Water is easier to reduce than Na^+ ($E°$ for Na^+/Na is very negative, −2.71 V). Therefore H_2O/H^+ is reduced to H_2 at the cathode instead of Na^+ being reduced to Na.

f. Calculate the oxidation number of Cr in: (i) $CrCl_3$ (ii) $Cr_2(SO_4)_3$ (iii) $Cr_2O_7^{2-}$.

(i) $x + 3(-1) = 0 \rightarrow x = +3$. (ii) $2x + 3(-2) = 0 \rightarrow x = +3$. (iii) $2x + 7(-2) = -2 \rightarrow 2x = +12 \rightarrow x = +6$.

g. Reactivity order K > Mg > Zn > Fe > Cu. Write balanced equations (if any) for: (i) Cu added to $MgSO_4$ solution (ii) Fe added to dilute HCl.

(i) No reaction: Cu is less reactive than Mg and cannot displace Mg^{2+}.

(ii) $Fe + 2HCl \rightarrow FeCl_2 + H_2$ (Fe is above H and displaces hydrogen).

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h. Why can some metals higher in the activity series displace hydrogen from acids while others lower cannot?

Metals above hydrogen lose electrons more easily than H_2, so they reduce H^+ to H_2. Metals below hydrogen (Cu, Ag, Pt) are less easily oxidized and cannot reduce H^+, so they do not displace hydrogen.

i. Calculate the number of faradays required to deposit 108 g Ag^+, 63.5 g Cu^{2+} and 27 g Al^{3+}.

$Ag^+ + e^- \rightarrow Ag$: 108 g = 1 mol Ag → 1 F. $Cu^{2+} + 2e^- \rightarrow Cu$: 63.5 g = 1 mol Cu → 2 F. $Al^{3+} + 3e^- \rightarrow Al$: 27 g = 1 mol Al → 3 F.

j. In electrolysis of $AgNO_3$, 0.500 A for 30.0 min deposits 0.503 g Ag. M(Ag) = 107.87, charge +1. Calculate Avogadro's number from this data.

Q = It = 0.500 × 1800 = 900 C. n(Ag) = 0.503/107.87 = 0.004663 mol. Since $Ag^+ + e^- \rightarrow Ag$, moles of electrons = 0.004663. Experimental F = Q/n = 900/0.004663 = 1.93×10^5 C mol^1. $N_a = F/e = (1.93 \times 10)/(1.602 \times 10^{-19}) = 1.20 \times 10^{24}$ mol^{-1}.

The printed data give $N_a \approx 1.2 \times 10^{24}$, about twice the accepted 6.02×10^{23}; the deposited mass (0.503 g) is roughly half the theoretical value (~1.01 g) for 900 C, indicating a data/printing inconsistency.

k. A cell has Ni^{2+}/Ni ($E° = -0.25$ V) and Co^{2+}/Co ($E° = -0.28$ V). (i) Identify anode/cathode. (ii) Overall reaction. (iii) E°cell.

(i) Co has the more negative E°, so Co is the anode (oxidized) and Ni is the cathode (reduced).

(ii) $Co(s) + Ni^{2+}(aq) \rightarrow Co^{2+}(aq) + Ni(s)$.

(iii) $E°cell = E°cathode - E°anode = (-0.25) - (-0.28) = +0.03$ V.

Descriptive / Long Questions

Q.3 How does electrode potential vary with concentration of an aqueous solution? Use the Nernst equation.

The Nernst equation relates electrode potential to ion concentration. For a reduction $M^{n+} + ne^- \rightarrow M$ at 298 K: $E = E° + (0.059/n) \log[M^{n+}]$.

Increasing the metal-ion concentration makes E more positive (greater tendency for reduction); decreasing it makes E more negative. For a general cell, $E = E° - (0.059/n) \log Q$, where Q is the reaction quotient.

Q.4 How can Avogadro's number be derived using an electrolytic cell?

Pass a known charge Q = It through an electrolytic cell and measure the mass of substance deposited. Calculate moles deposited (n = m/M) and the moles of electrons from the electrode equation. The charge per mole of electrons gives the Faraday constant, F = Q/(moles of e^-). Since $F = N_a \times e$ (where $e = 1.602 \times 10^{-19}$ C), $N_a = F/e$.

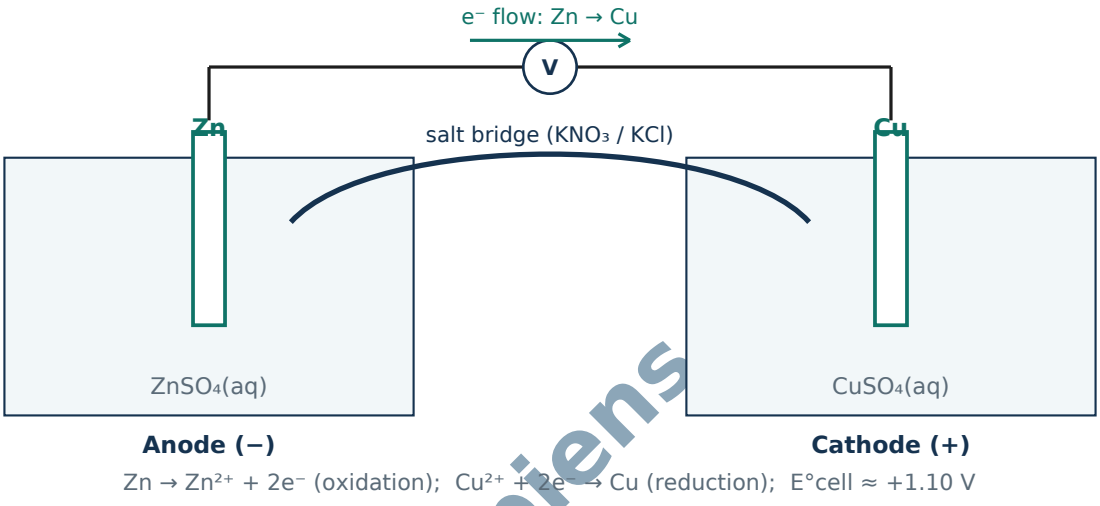
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Q.5 Describe the construction and working principle of the Zn-Cu galvanic cell.

The cell has a Zn rod in $ZnSO_4$ solution (anode) and a Cu rod in $CuSO_4$ solution (cathode), connected by a wire and a salt bridge.

At the anode: $Zn \rightarrow Zn^{2+} + 2e^-$ (oxidation). At the cathode: $Cu^{2+} + 2e^- \rightarrow Cu$ (reduction). Electrons flow from Zn to Cu through the wire; the salt bridge maintains neutrality. Overall: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$; $E°cell \approx +1.10$ V.

Zn–Cu galvanic (Daniell) cell

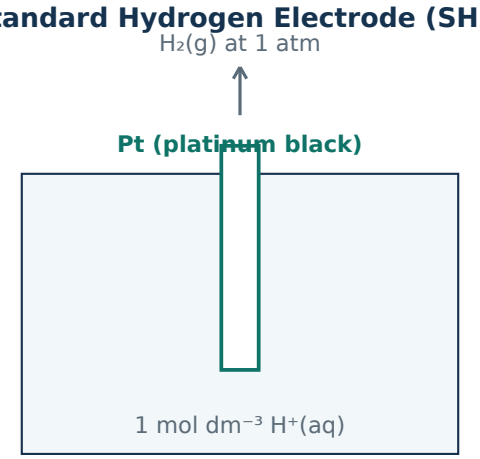


Q.6 What is meant by Standard Hydrogen Electrode (SHE)? How is it used to measure the electrode potential of another electrode?

SHE consists of a platinum black electrode in contact with H_2 gas at 1 atm and H^+ ions at 1 mol dm^{-3}, at 298 K. Its standard reduction potential is assigned 0.00 V: $2H^+ + 2e^- \rightleftharpoons H_2$.

To measure another electrode, it is connected to SHE to form a cell. The measured cell voltage gives the electrode potential of the other half-cell ($E°cell = E°cathode - E°anode$). If the other electrode is the cathode, its E° is positive; if it is the anode, its E° is negative.

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$2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$; $E^\circ = 0.00$ V at 298 K

Numerical Problems (fully worked)

Q.7 Calculate the electrode potential for a zinc electrode in 0.010 mol dm^{-3} $ZnSO_4$ at 298 K. $E^\circ(Zn^{2+}/Zn) = -0.76$ V. R = 8.31 J K^{-1} mol^{-1}, F = 96500 C mol^{-1}.

Given: $[Zn^{2+}] = 0.010$ M; T = 298 K; n = 2; $E^\circ = -0.76$ V

Formula: $E = E^\circ + (RT/nF) \ln[Zn^{2+}]$

$RT/nF = (8.31 \times 298)/(2 \times 96500) = 0.01284$ V.

$E = -0.76 + 0.01284 \times \ln(0.010) = -0.76 + 0.01284 \times (-4.605) = -0.76 - 0.0591 = -0.819$ V.

Answer: $E \approx -0.82$ V

Q.8 A constant current of 2.00 A is passed through $CuSO_4$ for 30.0 min. Calculate the mass of copper deposited. M(Cu) = 63.5 g mol^{-1}, F = 96500 C mol^{-1}.

Given: I = 2.00 A; t = 30.0 min = 1800 s; n = 2 ($Cu^{2+} + 2e^- \rightarrow Cu$)

Formula: $Q = It$; moles Cu = $Q/(nF)$; mass = moles × M

$Q = 2.00 \times 1800 = 3600$ C.

moles Cu = $3600/(2 \times 96500) = 0.01865$ mol.

mass = $0.01865 \times 63.5 = 1.18$ g.

Answer: Mass of Cu ≈ 1.18 g

Q.9 A galvanic cell consists of SHE and a Ni^{2+}/Ni half-cell. The measured cell potential at 298 K is 0.25 V and the nickel electrode is the negative terminal. (a) Overall reaction. (b) E° of Ni^{2+}/Ni. (c) Anode and cathode.

Given: $E^\circ_{cell} = 0.25$ V; Ni is the negative terminal (anode); SHE $E^\circ = 0.00$ V

Formula: $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

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(c) Ni is the anode (negative terminal, oxidation); SHE is the cathode (reduction).

(a) $Ni(s) + 2H^+(aq) \rightarrow Ni^{2+}(aq) + H_2(g)$.

(b) $0.25 = 0.00 - E°(Ni^{2+}/Ni) \rightarrow E°(Ni^{2+}/Ni) = -0.25$ V.

Answer: (a) $Ni + 2H^+ \rightarrow Ni^{2+} + H_2$; (b) $E°(Ni^{2+}/Ni) = -0.25$ V; (c) Ni = anode, SHE = cathode

Quick Check Solutions (in-chapter practice boxes)

Quick Check 10.1 (book p. 211)

a. Define oxidation and reduction in terms of electron transfer, with examples.

Oxidation is loss of electrons (e.g. $Na \rightarrow Na^+ + e$). Reduction is gain of electrons (e.g. $Cl_2 + 2e^- \rightarrow 2Cl^-$).

b. Define oxidation and reduction in terms of oxidation number, with examples.

Oxidation is an increase in oxidation number (e.g. $Fe^{2+} \rightarrow Fe^{3+}$). Reduction is a decrease (e.g. $Cu^{2+} \rightarrow Cu$).

c. Determine oxidation numbers: (i) O in Na_2O (ii) I in ICl_3 (iii) N in NO_3^- (iv) Cr in $K_2Cr_2O_7$ (v) Mn in $KMnO_4$.

(i) −2. (ii) +3. (iii) +5. (iv) +6. (v) +7.

Quick Check 10.2 (book p. 212)

a. Identify oxidized/reduced species and oxidizing/reducing agents: (i) $Fe_2O_3 + CO \rightarrow Fe + CO_2$ (ii) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$.

(i) Fe is reduced (+3 → 0); C is oxidized (+2 → +4). Fe_2O_3 is the oxidizing agent; CO is the reducing agent. (ii) C is oxidized (−4 → +4); O is reduced (0 → −2). CH_4 is the reducing agent; O_2 is the oxidizing agent.

b. Identify the species undergoing both oxidation and reduction: (i) $2Na_2O_2 + 2H_2O \rightarrow 4NaOH + O_2$ (ii) $P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + 3H_2PO_2^-$.

(i) Oxygen in Na_2O_2 (−1) is both reduced to −2 (in NaOH) and oxidized to 0 (in O_2) — disproportionation. (ii) Phosphorus (0) is both reduced to −3 (PH_3) and oxidized to +1 ($H_2PO_2^-$) — disproportionation.

Quick Check 10.3 (book p. 214)

i. Balance $Fe_2O_3 + CO \rightarrow Fe + CO_2$ by oxidation number method.

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$.

ii. Balance $MnO_2 + HCl \rightarrow MnCl_2 + H_2O + Cl_2$.

$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$.

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iii. Balance $Fe^{2+} + MnO_4^- \rightarrow Fe^{3+} + Mn^{2+}$ (acidic).

$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$.

iv. Balance $K_2Cr_2O_7 + FeSO_4 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + Fe_2(SO_4)_3 + K_2SO_4 + H_2O$.

$K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + K_2SO_4 + 7H_2O$.

Quick Check 10.4 (book p. 215)

i. Ions present in molten $CuBr_2$.

Cu^{2+} and Br^-.

ii. Electrode on which copper forms.

The cathode (negative electrode).

iii. Half-equation for copper formation.

$Cu^{2+} + 2e^- \rightarrow Cu$.

iv. Half-equation for bromine formation.

$2Br^- \rightarrow Br_2 + 2e^-$ (at the anode).

Quick Check 10.5 (book p. 219)

a. Mass of Ag deposited from $AgNO_3$ in 35 min at 0.18 A. (Ag = 108, F = 96500).

$Q = It = 0.18 \times 2100 = 378$ C. $n(Ag) = Q/F = 378/96500 = 0.00392$ mol. Mass = $0.00392 \times 108 = 0.423$ g.

b(i). Charge passed when 1.04 A flows for 6.00 min.

$Q = 1.04 \times 360 = 374.4$ C.

b(ii). Charge required to liberate 1 mole of H_2 gas.

$2H^+ + 2e^- \rightarrow H_2$ requires 2 mol electrons = $2F = 1.93 \times 10^5$ C. (From the data: 41.5 cm^3 H_2 = 0.00185 mol; experimental charge per mole ≈ 2.0×10^5 C, close to 2F.)

Quick Check 10.6 (book p. 224)

a. Function of the salt bridge in a voltaic cell.

It completes the circuit and maintains electrical neutrality by allowing ions to migrate between the half-cells, without mixing the solutions.

b. Calculate the voltage of a cell with iron and copper electrodes. $E°(Fe^{2+}/Fe) = -0.44$ V; $E°(Cu^{2+}/Cu) = +0.34$ V.

Fe is the anode, Cu the cathode. $E°cell = E°cathode - E°anode = 0.34 - (-0.44) = +0.78$ V.

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Quick Check 10.7 (book p. 226)

a(i). Explain using E° values: chlorine can oxidize bromide or iodide ions.

$E°(Cl_2/Cl^-)$ = +1.36 V is more positive than $E°(Br_2/Br^-)$ = +1.07 V and $E°(I_2/I^-)$ = +0.54 V, so Cl_2 is a stronger oxidizing agent and can oxidize Br^- to Br_2 and I^- to I_2.

a(ii). Why bromine does not react with chloride ions.

Br_2 is a weaker oxidizing agent than Cl_2 (E° +1.07 V < +1.36 V), so it cannot oxidize Cl^- to Cl_2.

Quick Check 10.8 (book p. 230)

a. State the Nernst equation and its significance.

$E = E° - (RT/nF) \ln Q$ (at 298 K: $E = E° - (0.059/n) \log Q$). It relates electrode/cell potential to ion concentrations, allowing prediction of potential under non-standard conditions.

b. Effect of ion concentration on electrode potential.

Increasing the concentration of the oxidized (metal ion) species makes the reduction potential more positive; decreasing it makes the potential more negative.

c(i). Strongest and least oxidizing agent among Ag^+, Cr^{2+}, Fe^{2+}. E°: Ag^+/Ag +0.80, Cr^{2+}/Cr −0.91, Fe^{2+}/Fe −0.44 V.

Strongest oxidizing agent: Ag^+ (most positive E°). Least oxidizing: Cr^{2+} (most negative E°).

c(ii). Arrange Ag, Cr, Fe in increasing order of reducing power.

Ag < Fe < Cr (more negative E° means the metal is more easily oxidized and is a stronger reducing agent).

d. Electrode potential of Cu in 0.01 M Cu^{2+}. E° = +0.34 V.

$E = E° + (0.059/2) \log[Cu^{2+}] = 0.34 + 0.0295 \log(0.01) = 0.34 - 0.059 = +0.28$ V.

e. Electrode potential of Cl^-/Cl_2 at 25 °C if $[Cl^-]$ = 0.05 M. $Cl_2 + 2e^- \rightarrow 2Cl^-$, E° = 1.36 V.

$E = E° - (0.059/2) \log([Cl^-]^2/pCl_2)$. Taking pCl_2 = 1 atm: $E = 1.36 - 0.0295 \log(0.05)^2 = 1.36 + 0.077 = +1.44$ V.

Quick Check 10.6(b): the printed E° for Cu^{2+}/Cu is shown as −0.44 V (same as Fe); the standard value is +0.34 V, used here.

Textbook checks flagged in this chapter

- p. 232, MCQ V: both (c) and (d) are reduction half-reactions.
- p. 234, Q.2(j): the printed data give $N_a \approx 1.2 \times 10^{24}$ (about twice the accepted value); the deposited mass appears inconsistent with the charge passed.

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